1. **What is your outcome variable?**

- Our outcome is **denied**, a binary variable built from the data's action_taken

> *We recoded action_taken into a binary outcome **accepted** vs **denied** after dropping loans, withdrawals, incomplete and prepapprovals*

2. **How well does it measure the outcome you are interested in?**

It measures the lender's final decision on applications that reach a decision. It equals **1** when the application was denied (action_taken = 3) and 0 when it was **approved**, whether the loan originated (action_taken = 1) or approved but not accepted (action_taken = 2). We dropped withdrawn applications, files closed for incompleteness, purchased loans, and preapprovals **because they do not reflect a lender's credit decision - It was done by the applicant which doesn't reflect on the lender system(done by an underwriter) we are interested in**. Of the 10,513 applications in our sample, 1,476 were denied, for an overall denial rate of 14.0%.

It is recorded for nearly every application under HMDA, so it is consistent across lenders. However, it is an imperfect measure in four ways:

- Applicants who expect to be denied may withdraw, and some weak applications are closed as incomplete instead of denied. Because we drop these cases, our measured gaps may understate true differences in access.
- It does not capture loan pricing. An applicant who is approved at a higher interest rate counts as a success, even though Lewis-Faupel and Tenev (2024) show pricing is a second channel of disparity.
- It does not capture people who were discouraged from applying at all.
- The data does not include credit scores, so we cannot fully separate creditworthiness from other explanations.

3. **How does it relate to your expectations?**

The unadjusted denial rates largely match our expectations. Non-Hispanic White applicants have the lowest denial rate (10.9%). Asian applicants (15.0%) are somewhat higher, and Hispanic (21.8%) and Black (24.7%) applicants are denied at roughly twice the White rate.

Denial rates also rise with the minority share of the census tract, from 11.0% in tracts under 20% minority to 25.4% in tracts that are 80 to 100% minority.

Our next step is to test whether these gaps shrink but persist once we control for income, loan amount, DTI, loan type, lender, and metro area.

4. **What are your key explanatory variables?**

- Applicant race and ethnicity.
- Tract minority population percentage

Our control variables are applicant income (log-transformed), loan amount (log-transformed), debt-to-income ratio (grouped into bands), loan type, lender (lei), metro area (derived_msa-md), and applicant age.

5. Summary Statistics

	count	missing	mean	std	min	25%	50%	75%
income	10118.0	395	290.7	483.3	-719.0	106.0	173.0	317.0
loan_amount	10513.0	0	669449.7	692452.2	15000.0	265000.0	485000.0	795000.0
tract_minority_population_percent	10513.0	0	42.9	25.6	4.2	22.5	36.2	60.2
tract_to_msa_income_percentage	10513.0	0	143.6	72.1	0.0	92.0	122.0	181.0
tract_population	10513.0	0	4846.7	1929.5	44.0	3497.0	4628.0	5950.0

== race_eth ==

race_eth	n	denial_rate	ci_low	ci_high	share
White	5233	0.109	0.100	0.117	0.498
Asian	2255	0.150	0.135	0.165	0.214
Hispanic (any race)	934	0.218	0.192	0.245	0.089
Race Not Available	879	0.142	0.119	0.165	0.084
Black or African American	726	0.247	0.215	0.278	0.069
Joint	436	0.112	0.083	0.142	0.041
American Indian or Alaska Native	16	0.188	0.000	0.379	0.002
2 or more minority races	16	0.250	0.038	0.462	0.002
Native Hawaiian or Other Pacific Islander	11	0.273	0.010	0.536	0.001
Free Form Text Only	7	0.286	0.000	0.620	0.001

== derived_race ==

derived_race	n	denial_rate	ci_low	ci_high	share
White	5841	0.118	0.110	0.126	0.556
Asian	2271	0.151	0.136	0.166	0.216
Race Not Available	1056	0.166	0.143	0.188	0.100
Black or African American	808	0.246	0.217	0.276	0.077
Joint	452	0.115	0.086	0.144	0.043
American Indian or Alaska Native	28	0.214	0.062	0.366	0.003
Native Hawaiian or Other Pacific Islander	25	0.120	0.000	0.247	0.002
2 or more minority races	21	0.238	0.056	0.420	0.002
Free Form Text Only	11	0.273	0.010	0.536	0.001

```
== derived_ethnicity ==
                            n  denial_rate  ci_low  ci_high  share
derived_ethnicity
Not Hispanic or Latino   8408        0.132   0.125    0.139  0.800
Hispanic or Latino        934        0.218   0.192    0.245  0.089
Ethnicity Not Available   831        0.153   0.128    0.177  0.079
Joint                     326        0.095   0.063    0.127  0.031
Free Form Text Only        14        0.143   0.000    0.326  0.001

== lei ==
                         n  denial_rate  ci_low  ci_high  share
lei
7H6GLXDRUGQFU57RNE97  6643        0.121   0.113    0.129  0.632
KB1H1DSPRFMYMCUFXT09  2217        0.154   0.139    0.169  0.211
B4TYDEB6GKMZO031MB27  1653        0.201   0.182    0.221  0.157

== loan_type ==
               n  denial_rate  ci_low  ci_high  share
loan_type
1          10285        0.134   0.128    0.141  0.978
2            177        0.492   0.418    0.565  0.017
3             51        0.157   0.057    0.257  0.005

== dti_band ==
              n  denial_rate  ci_low  ci_high  share
dti_band
36%-49%    5276        0.105   0.097    0.113  0.502
20%-<30%   2111        0.088   0.076    0.100  0.201
30%-<36%   1946        0.089   0.076    0.102  0.185
<20%        605        0.116   0.090    0.141  0.058
>60%        288        0.969   0.949    0.989  0.027
50%-60%     264        0.746   0.694    0.799  0.025



  == tract_minority_band ==
                          n  denial_rate  ci_low  ci_high  share
  tract_minority_band
  20-<40%              3804        0.117   0.107    0.127  0.362
  40-<60%              2055        0.131   0.117    0.146  0.195
  0-<20%               2005        0.110   0.096    0.123  0.191
  80-100%              1362        0.254   0.231    0.277  0.130
  60-<80%              1287        0.152   0.132    0.171  0.122

  == age_band ==
               n  denial_rate  ci_low  ci_high  share
  age_band
  35-44     3573        0.127   0.116    0.137  0.340
  25-34     3305        0.121   0.110    0.132  0.314
  45-54     1777        0.167   0.149    0.184  0.169
  55-64     1029        0.172   0.149    0.195  0.098
  65-74      456        0.178   0.143    0.213  0.043
  <25        251        0.215   0.164    0.266  0.024
  >74        111        0.144   0.079    0.209  0.011
  Missing     11        0.091   0.000    0.261  0.001
```